

Optimized cutting plan

Saunacloud project (invoice #1026) · 56 frames remaining to produce

Material to purchase

Stock	Rip	2×4 needed	+12% margin	Use
Structural	×2 (longs + shorts)	24	27	Frame perimeter
Traverses	×4 (thin slats)	10	12	Interior slats
TOTAL 2×4		34	39	
1×6 boards	—	9	10	FS750 full-spectrum

Purchase list: 39 × 2×4 × 8 ft + 10 × 1×6 × 8 ft boards.

Existing stock used (in shop)

Piece	Qty on hand	Equivalent after rip ×2	Covers
20" (un-ripped)	10	20 ripped pieces of 20"	1/3 of 20" need
33" (un-ripped)	5	10 ripped pieces of 33"	1/2 of 33" need

The above stock is deducted from the total need before computing the purchase — savings of 5 × 2×4 compared to buying everything new.

Two stocks — rip strategy

- **Structural stock:** 2×4 ripped in half (~1½" thick) for the frame perimeter (2 longs + 2 shorts per frame).
- **Traverse stock:** 2×4 ripped in four (~½–¾" × 2") for the interior slats. One 2×4 yields 4 traverses side-by-side.

Length color legend

37"
 33"
 26"
 22"
 20"
 13"

Piece role per SKU (to produce)

SKU	Long ×2	Short ×2	Traverses
H2013 (20×13") — 20 frames	13"	20"	2 × 20"
H2026 (20×26") — 10 frames	26"	20"	4 × 20"
H2226 (22×26") — 6 frames	26"	22"	4 × 22"
H3313 (33×13") — 10 frames	33"	13"	2 × 13"
H3726 (37×26") — 10 frames	37"	26"	4 × 26"

Cut patterns — Structural stock (24 × 2×4 new)

Each 2×4 is ripped in half along the length, yielding 2 usable 96" lengths. Total: **47 lengths** = 24 × new 2×4 ripped. The existing stock (10 × 20" and 5 × 33") is ripped separately and used directly.

37"
 33"
 26"
 22"
 20"
 13"

P1 15× planks · 26" + 26" + 26" + 13" · waste 4.6"



P2 10× planks · 37" + 37" + 20" · waste 1.8"



P3 7× planks · 20" + 20" + 20" + 20" + 13" · waste 2.5"



P4 5× planks · 33" + 33" + 26" · waste 3.8"



P5 5× planks · 13" + 13" + 13" + 13" + 13" + 13" + 13" · waste 4.2"



P6 2× planks · 22" + 22" + 22" + 22" · waste 7.6"



P7 1× planks · 26" + 26" + 22" + 20" · waste 1.6"



P8 1× planks · 22" + 22" + 22" + 20" · waste 9.6"



P9 1× planks · 13" + 13" + 13" · waste 56.8"



Cut patterns — Traverse stock (10 × 2×4)

Each 2×4 is ripped in four along the length, yielding 4 usable 96" thin strips ($\frac{1}{2}$ – $\frac{3}{4}$ " × 2"). Total: **40 lengths** = 10 × 2×4.

PT1 13× planks · 26" + 26" + 26" + 13" · waste 4.6"



PT2 13× planks · 20" + 20" + 20" + 20" · waste 15.6"

20"	20"	20"	20"	15.6"
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PT3 6× planks · 20" + 20" + 20" + 20" + 13" · waste 2.5"

20"	20"	20"	20"	13"	
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PT4 5× planks · 22" + 22" + 22" + 22" · waste 7.6"

22"	22"	22"	22"	7.6"
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PT5 1× planks · 22" + 20" + 20" + 20" + 13" · waste 0.5"

22"	20"	20"	20"	13"	
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PT6 1× planks · 26" + 22" + 22" + 22" · waste 3.6"

26"	22"	22"	22"	3.6"
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PT7 1× planks · 20" · waste 76.0"

20"	76.0"
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Total purchase cost

Complete cost of the wood purchase run, including truck rental and gas.

Item	Qty	Unit price	Subtotal	With taxes
2×4 × 8 ft	39	\$17.95	\$700.05	\$804.88
1×6 × 8 ft	10	\$16.95	\$169.50	\$194.88
Wood subtotal			\$869.55	\$999.77
U-Haul rental	1	\$85.00	\$85.00	\$97.73
Gas	—	—	\$20.00	\$20.00
GRAND TOTAL			\$974.55	\$1117.49

Taxes: GST 5% + QST 9.975%. Gas is an estimate and does not include taxes at the pump.

Pickup logistics

- **Vehicle:** U-Haul cargo van or small truck (\$85 base rental).
- **Wood volume:** 39 × 2×4 (~1½" × 3½" × 8 ft) + 10 × 1×6 (~¾" × 5½" × 8 ft) — fits in a cargo van.
- **Recommended supplier:** Langevin (lumber, Montreal) for consistent quality, or Home Depot for convenience.
- **Gas budget:** \$20 covers ~40 km round trip within Montreal.